

PhysicalRisk Platform

Developer Guide

MKM Research Labs

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1 Executive Summary

PhysicalRisk is a flood risk assessment and Physical Risk Swap (PRS) pricing platform developed by MKM Research Labs. The platform generates synthetic portfolios of flood gauges, residential properties, and mortgages within a river catchment, then simulates storm events to compute property-level flood hazard curves, risk-adjusted mortgage valuations, and PRS contract pricing.

1.1 Capabilities

- **Synthetic portfolio generation** — configurable portfolios of gauges, properties, mortgages, storms, and counterparties.
- **Flood simulation** — storm-to-gauge response modelling with spatial interpolation to property level using inverse distance weighting.
- **Hazard curve construction** — GEV-fitted exceedance probabilities at gauge and property level.
- **PRS pricing** — CDS-equivalent quarterly pricing of Physical Risk Swaps with five-component basis waterfall analysis.
- **Mortgage pricing** — risk-adjusted mortgage valuation with affordability, LTV, and flood risk credit spread adjustments.
- **Interactive visualisation** — Folium-based map with tabbed analysis panels, context menus, flood animation, and PDF report generation.
- **REST API** — Flask server exposing all data and analytics via JSON endpoints.
- **PDF reporting** — automated property, gauge, risk portfolio, and PRS trade confirmation reports.

1.2 Architecture Overview

```

1 app.py (CLI)
2 |-- server      --> src/server.py --> src/routes/*.py (Flask API)
3 |-- port       --> src/port/src/*.py (generators)
4 |             |-- src/port/cdm/*.py (schemas)
5 |             |-- src/port/rand/thames/*.py (random modules)
6 |             |-- src/models/*.py (analytical models)
7 |-- visual     --> src/visual/ (Folium map + interactivity)
8 |-- check      --> dependency verification
9 |-- config     --> configuration display
10
11 config.py      --> PortfolioConfig (singleton, paths, catchment)
12 data/
13 |-- input/thames/ (generated portfolios + timeseries)
14 |-- output/       (PDF reports, PRS trades)
15 |-- results/      (interactive visualisation HTML)

```

Listing 1: Platform architecture

2 Installation and Setup

2.1 Prerequisites

- Python 3.13 or later
- GDAL system library (for geospatial operations)
- QuantLib (optional, for CDS pricing — analytical fallback available)
- Node.js and npm (optional, for JavaScript tests)
- L^AT_EX distribution (optional, for building documentation PDFs)

2.2 Installation

```

1 # Clone repository
2 git clone <repository-url> PhysicalRisk
3 cd PhysicalRisk
4
5 # Install Python dependencies
6 pip install -r requirements.txt
7
8 # Verify installation
9 python app.py check

```

Listing 2: Installation steps

The `check` command verifies all required packages: `flask`, `flask_cors`, `folium`, `pandas`, `numpy`, `geopandas`, `reportlab`, `scipy`, `sklearn`, `rasterio`, `geopy`, `shapely`, `matplotlib`, `seaborn`. Each is reported as OK or MISSING.

2.3 Environment Variables

All settings have sensible defaults but can be overridden:

Table 1: Environment variable configuration

Variable	Default	Description
MKM_SERVER_HOST	127.0.0.1	Server bind address
MKM_SERVER_PORT	5013	Server port
MKM_DEBUG	false	Flask debug mode
MKM_CATCHMENT	thames	Active catchment ID
MKM_PROJECT_ROOT	Auto-detected	Override project root
MKM_INPUT_DIR	data/input	Override input directory
MKM_OUTPUT_DIR	data/output	Override output directory
MKM_RESULTS_DIR	data/results	Override results directory

3 Configuration

3.1 Configuration System

The platform uses two configuration classes in `config.py`:

- **Config** — simple class with server settings as class attributes. Used by Flask for host/port/debug/CORS.
- **PortfolioConfig** — singleton managing catchment, directory paths, and module loading. The global config instance is available via `from config import config`.

```

1 $ python app.py config
2 Project root:    /path/to/PhysicalRisk
3 Catchment:      thames
4 Input dir:      /path/to/PhysicalRisk/data/input/thames
5 Output dir:     /path/to/PhysicalRisk/data/output
6 Reports dir:    /path/to/PhysicalRisk/data/output
7 Results dir:    /path/to/PhysicalRisk/data/results
8 Server URL:     http://127.0.0.1:5013
9 Debug:         False

```

Listing 3: View current configuration

3.2 Directory Structure

```

1 PhysicalRisk/
2   app.py           CLI entry point
3   config.py        Configuration (adds src/ to sys.path)
4   requirements.txt  Python dependencies
5   Dockerfile       Container image
6   docker-compose.yml Container orchestration
7   src/
8     server.py      Flask application factory
9     routes/        API endpoint blueprints
10    visual/        Map visualisation + interactivity
11    models/        Analytical models (centralised)
12    loaders/       Data loading classes
13    port/          Portfolio generators + CDM + random
14    reports/       PDF report generators
15    static/        Web assets (HTML, JS)
16    catch/         Catchment definitions
17    floodts/       Flood timeseries utilities
18  data/
19    input/<catchment>/ Generated portfolio data
20    output/        PDF reports and PRS trades
21    results/       Interactive visualisation HTML
22  tests/          Python and JavaScript tests
23  docs/           Documentation (LaTeX)

```

Listing 4: Project directory layout

3.3 Catchment System

Each catchment defines its geographic region in `data/catch/<id>.py`, inheriting from `BaseCatchment`:

- `CATCHMENT_ID` — unique identifier (e.g. `thames`)
- `GAUGE_POINTS` — list of (lat, lon, elevation_m) tuples
- `AREAS` — region/borough names for labelling

- `CENTER_LAT`, `CENTER_LON` — map centre coordinates
- `generate_synchronized_locations(count)` — produce property locations consistent with catchment geography

The Thames catchment defines 40 gauge points along the river from Richmond to Tilbury (elevations 4–11m) across 30 London boroughs. Additional catchments (Rhine, Mississippi, Danube, etc.) are defined but not yet populated with random modules.

4 CLI Reference

All commands are invoked via `python app.py <command> [options]`.

4.1 server — Start Web Server

```
1 python app.py server [--host HOST] [--port PORT] [--debug]
```

Starts the Flask web server. Registers all API route blueprints and serves the interactive visualisation. Default: `http://127.0.0.1:5013`.

4.2 port — Generate Portfolio Data

```
1 python app.py port [SEGMENTS] [OPTIONS]
```

Generates synthetic portfolio data. Segments must be run in dependency order (see Section 5).

4.2.1 Segment Flags

Table 2: Portfolio generation segment flags

Flag	Short	Output	Description
<code>--gauges</code>	<code>--ga</code>	<code>gauge.json</code>	Flood gauge stations
<code>--properties</code>	<code>--pr</code>	<code>property.json</code>	Property portfolio
<code>--mortgages</code>	<code>--mo</code>	<code>mortgage.json</code>	Mortgage data
<code>--gaugets</code>	<code>--gt</code>	<code>gaugets/*.json</code>	Gauge flood simulations
<code>--storms</code>	<code>--st</code>	<code>storms.json</code>	Storm scenarios
<code>--gaugehd</code>	<code>--hd</code>	<code>gaugehd/*.json</code>	Historical daily data
<code>--hazard</code>	<code>--hz</code>	<code>gaugehc.json</code>	Gauge hazard curves
<code>--propertyts</code>	<code>--pt</code>	<code>propertyts/*.json</code>	Property flood events
<code>--propertyhc</code>	<code>--phc</code>	<code>propertyhc.json</code>	Property hazard + PRS
<code>--counterparties</code>	<code>--ctpy</code>	<code>counterparty.json</code>	Swap counterparties
<code>--all</code>	<code>-a</code>	All above	Run all segments

4.2.2 Global Options

Table 3: Portfolio generation options

Flag	Short	Default	Description
<code>--num-properties</code>	<code>-np</code>	200	Number of properties
<code>--num-gauges</code>	<code>-ng</code>	40	Number of gauges
<code>--num-storms</code>	<code>-ns</code>	10,000	Number of storm scenarios
<code>--simulation-hours</code>		60	Flood simulation duration
<code>--history-years</code>	<code>-hy</code>	50	Historical data years
<code>--tail-weight</code>	<code>-tw</code>	2.0	Storm intensity tail weight
<code>--distribution</code>	<code>-d</code>	gev	Hazard distribution (gev gumbel)
<code>--no-backup</code>		false	Skip automatic data backup
<code>--verbose</code>	<code>-v</code>	false	Verbose output

4.3 visual — Generate Interactive Map

```
1 python app.py visual [--no-browser]
```

Generates an interactive Folium map at `data/results/interactive_visualization.html` and opens it in the default browser (unless `--no-browser` is specified).

4.4 check — Verify Dependencies

```
1 python app.py check
```

Reports the status of each required Python package. Exits with code 1 if any packages are missing.

4.5 config — Show Configuration

```
1 python app.py config
```

Displays current project root, catchment, directory paths, server URL, and debug mode.

5 Data Pipeline

5.1 Generation Sequence

Segments must be generated in dependency order. Each segment reads from the output of earlier segments:

```
1 gauges --> properties --> mortgages
2     |
3 gauges --> gaugets --> storms --> gaugehd --> hazard
4     |                                     |
5     +-----> propertyts -----> propertyhc
6                                     |
7                                counterparties -----+
```

Listing 5: Data pipeline dependency chain

Running `python app.py port --all` executes all segments in the correct order. When regenerating specific segments, ensure upstream dependencies exist.

5.2 Data Backup

Before generating new data, the existing input directory is automatically backed up to `audit/<catchment>_YYYYMM`. This preserves an audit trail of all data changes. Use `--no-backup` to skip.

5.3 Segment Details

5.3.1 Gauges

Generates flood gauge stations with location, elevation, and flood stage thresholds (Alert, Warning, Severe). Uses catchment-defined gauge points. Each gauge includes sensor metadata (type, owner, installation date) and historical statistics.

Output: `gauge.json` containing `flood_gauges[]` array.

5.3.2 Properties

Generates residential properties with location, construction details, and multi-factor valuation. Property value is computed as:

$$V = \text{area} \times \frac{\text{price}}{\text{sqm}} \times f_{\text{location}} \times f_{\text{age}} \times f_{\text{condition}} \times f_{\text{flood}} \times f_{\text{EPC}} \times f_{\text{proximity}}$$

The critical `FloorLevelMeters` field determines whether a property floods — only water above this threshold (relative to ground elevation) counts as internal flooding.

Output: `property.json` containing `properties[]` array.

5.3.3 Mortgages

Generates mortgage instruments linked to properties by `PropertyID`. Includes loan principal, interest rate, term, borrower income, and computes affordability ratio and LTV. Each mortgage carries a `FloodRiskCategory` (Very Low through Very High).

Output: `mortgage.json` containing `mortgages[]` array.

5.3.4 Gauge Time Series (gaugets)

Simulates 60-hour flood events for each gauge. Produces hourly water level readings showing baseline, storm rise, peak, and recession. Also generates storm response records linking storms to gauge-level peak levels.

Output: `gaugets/GAUGE-xxx.json` per gauge, containing `flood_simulation` (hourly readings) and `storm_responses`.

5.3.5 Storms

Generates storm scenarios using a Hybrid Normal-Pareto intensity distribution. Most storms are benign; rare events cause significant gauge responses. Each storm has spatial extent, track, duration, and intensity parameters.

Output: `storms.json` containing `storms[]` array (default 10,000 events, ~6 MB).

5.3.6 Gauge Historical Data (`gaugehd`)

Generates daily water level records spanning the configured number of years (default 50). Computes percentiles (p5–p99), annual maxima, and flood exceedance counts. Can also parse NRFA GDF CSV format.

Output: `gaugehd/GAUGE-xxx.json` per gauge, containing `daily_observations[]` and `statistics`.

5.3.7 Hazard Curves

Reads storms and gauge time series, simulates storm-to-gauge responses, extracts peak levels, and fits a GEV (Generalised Extreme Value) distribution. Computes exceedance probabilities at return periods (1, 5, 10, 25, 50, 100 years) and prices gauge-level PRS contracts.

Output: `gaugehc.json` containing per-gauge GEV parameters, hazard curve points, return period levels, and PRS term structures.

5.3.8 Property Time Series (`propertyts`)

For each property, identifies the three nearest gauges and interpolates flood depth using inverse distance weighting ($1/d^2$). Applies elevation differentials and floor level thresholds. Uses Manning's equation for travel time and attenuation. Computes damage ratio via depth-damage curve.

Output: `propertyts/PROP-xxx.json` per property, containing `flood_events[]` with depth, damage ratio, arrival time, and hourly readings. Also `portfolio_flood_summary.json` with aggregate statistics.

5.3.9 Property Hazard Curves (`propertyhc`)

Fits GEV to property flood depths (requires ≥ 3 events). Computes exceedance probabilities at three depth thresholds ($>0\text{m}$, $>0.5\text{m}$, $>1.0\text{m}$). Prices property-level PRS via analytical CDS model. Computes five-component basis waterfall (distance, elevation, terrain, composition, model uncertainty).

Output: `propertyhc.json` containing per-property GEV parameters, hazard curves, PRS spreads, and basis analysis.

5.3.10 Counterparties

Generates synthetic counterparty entities (banks, insurers, reinsurers) with credit ratings, exposure limits, and collateral requirements. Used for PRS trade management.

Output: `counterparty.json`.

5.4 Common Data Model (CDM)

All entities follow standardised schemas defined in `src/port/cdm/`:

Table 4: CDM schema registry

CDM Class	File	Key Sections
FloodGaugeCDM	gauge.py	Header, Location, FloodStages, SensorDetails
PropertyCDM	property.py	Header, Location, Valuation, Construction
MortgageCDM	mortgage.py	Header, FinancialTerms, Borrower, Risk
StormEventCDM	storm.py	StormHeader, Intensity, Spatial, Timing
StormTimeSeriesCDM	stormts.py	TimeseriesHeader, Readings
PhysicalRiskSwapCDM	prs.py	SwapHeader, Pricing, Basis
CounterpartyCDM	ctpy.py	EntityHeader, Exposure, Collateral

All CDMs inherit from `BaseCDM` and provide `validate(data)`, `create_mapping(data)`, and a `schema` property.

6 Web Server and API Reference

6.1 Starting the Server

```
1 python app.py server --port 5013 --debug
```

The server registers route blueprints and serves both the API and the interactive visualisation.

6.2 Blueprint Organisation

Table 5: API route blueprints

Blueprint	Prefix	Purpose
health_bp	/	Health check, root route
catchment_bp	/	Catchment selector page
visualization_bp	/	Interactive map generation
properties_bp	/api/v1	Property endpoints
gauges_bp	/api/v1	Gauge endpoints
propertyts_bp	/api/v1	Property flood timeseries
propertyhc_bp	/api/v1	Property hazard curves
counterparty_bp	/api/v1	Counterparty data
prs_bp	/api/v1	PRS trade management

6.3 Response Format

All API responses use a consistent JSON envelope:

```
1 // Success
2 {"status": "success", "data": {...}}
3
4 // Error
5 {"status": "error", "message": "Description of error"}
```

Listing 6: Standard API response format

HTTP status codes: 200 (success), 400 (bad request), 404 (not found), 500 (server error), 503 (missing dependencies).

6.4 Health and System Endpoints

Table 6: Health and system endpoints

Method	Path	Description
GET	/ or /health	Basic health status
GET	/health/detailed	Detailed health with file checks
GET	/select-catchment	Catchment selection HTML page
POST	/api/set-catchment	Set active catchment
GET	/visualization	Generate and serve interactive map

6.5 Property Endpoints

Table 7: Property API endpoints

Method	Path	Description
GET	/api/v1/properties	List all properties
GET	/api/v1/properties/<id>	Single property details
GET	/api/v1/properties/<id>/mortgage	Mortgage for property
POST	/api/v1/properties/report	Generate property PDF
POST	/api/v1/properties/mortgage-report	Generate mortgage PDF

6.6 Gauge Endpoints

Table 8: Gauge API endpoints

Method	Path	Description
GET	/api/v1/gauges	List all gauges
GET	/api/v1/gauges/<id>	Single gauge details
GET	/api/v1/gauges/<id>/history	Historical daily data
GET	/api/v1/gauges/<id>/timeseries	Flood simulation readings
GET	/api/v1/gauges/<id>/storms	Storm scenario analysis
GET	/api/v1/gauges/<id>/hazard	Hazard curve data
GET	/api/v1/gauges/<id>/statistics	Reading statistics
POST	/api/v1/gauges/report	Generate gauge PDF
POST	/api/v1/price_prs	Price a PRS contract

6.7 Property Flood Timeseries Endpoints

Table 9: Property flood timeseries endpoints

Method	Path	Description
GET	/api/v1/propertyts/summary	Portfolio flood summary
GET	/api/v1/properties/<id>/floods	Flood events for property
GET	/api/v1/properties/<id>/storms	Storm analysis with readings
GET	/api/v1/propertyts/storms	Storms causing property floods
GET	/api/v1/propertyts/animate/<storm_id>	Animation frames (60 hours)
GET	/api/v1/propertyts/animate/composite	Worst-case composite animation

6.8 Property Hazard Curve Endpoints

Table 10: Property hazard curve endpoints

Method	Path	Description
GET	/api/v1/propertyhc/summary	Portfolio hazard summary
GET	/api/v1/properties/<id>/hazard	Property hazard + PRS pricing
GET	/api/v1/propertyhc/basis	Basis analysis table

6.9 Counterparty and PRS Endpoints

Table 11: Counterparty and PRS trade endpoints

Method	Path	Description
GET	/api/v1/counterparties	List counterparties
GET	/api/v1/counterparties/<id>	Counterparty details
POST	/api/v1/prs/commit	Commit a PRS trade
GET	/api/v1/prs/trades	List committed trades
GET	/api/v1/prs/trades/<swap_id>/pdf	Download trade PDF

7 Interactive Visualisation

7.1 Map Generation

```
1 python app.py visual
```

Generates an interactive Folium HTML map with all portfolio data overlaid. The map includes tile layer controls, measurement tools, fullscreen toggle, and a scale indicator.

7.2 Map Layers

7.2.1 Gauge Layer

Gauge markers use water-droplet icons colour-coded by operational status (green = operational, orange = maintenance, red = offline). Tooltips show gauge ID and flood frequency. Click popups display

equipment details, sensor data, flood thresholds (alert/warning/severe levels), and historical context. A RAG indicator shows flood frequency risk.

7.2.2 Property Layer

Property markers use home icons colour-coded by flood frequency from simulated storms (red = high, orange = medium, green = low). Mortgaged properties show a bank icon variant. Click popups display property details, elevation and flood zone information, and mortgage details where applicable.

7.2.3 Mortgage Layer

Optional overlay showing translucent circles around mortgaged properties. Circle size correlates with loan amount; colour indicates flood risk level. High-leverage properties ($LTV > 80\%$) show a filled indicator.

7.3 Context Menus

Right-clicking a marker opens a context menu with actions:

Property context menu:

1. Property Details — view full property information
2. Mortgage Details — open mortgage detail panel
3. View Storm Scenarios — open Property Storm Analysis panel
4. Physical Risk Swap — open hazard curve and PRS pricing panel
5. Generate Property Report — create and view PDF report
6. Generate Mortgage Report — create mortgage-focused PDF

Gauge context menu:

1. View History — show historical water level chart
2. View Storm Scenarios — open Gauge Storm Analysis panel
3. Physical Risk Swap — open gauge hazard curve and PRS pricing
4. Generate Gauge Report — create and view PDF report

7.4 Interactive Panels

7.4.1 Property Storm Analysis

Five-tab panel (780×560px) loaded from `/api/v1/properties/{id}/storms`:

1. **Distribution** — flood depth histogram (0.2m bins) with event count, mean/max depth, gauge-to-property conversion rate.
2. **Flood Timeline** — 60-hour hydrograph for selected storm showing water surface elevation, property floor threshold, gauge reference lines, and red shading where water exceeds floor level. Storm selector dropdown. Stats: peak above floor, damage ratio, arrival/peak hours, flooded duration.
3. **Worst Storms** — horizontal bar chart of top 20 storms by flood depth, colour-coded by severity. Click a bar to switch to Timeline tab for that storm.
4. **Flood History** — scrollable table of all flood events (storm ID, depth, damage %) alongside a bar chart. Click rows to view timelines.

5. **Mortgage Impact** — stacked bar chart showing retained property value (blue) and damage loss (red) per storm. Outstanding balance shown as dashed horizontal line. Summary: property value, outstanding balance, current LTV, worst loss, worst post-damage LTV, negative equity count.

7.4.2 Gauge Storm Analysis

Three-tab panel (780×560px) loaded from `/api/v1/gauges/{id}/storms`:

1. **Distribution** — storm peak level histogram.
2. **Flood Timeline** — water level hydrograph with flood stage thresholds (alert, warning, severe).
3. **Worst Storms** — top 20 storms by peak level.

7.4.3 Property Hazard Curve and PRS Pricing

Four-tab panel (1100×750px) loaded from `/api/v1/properties/{id}/hazard`:

1. **Hazard Curve** — log-log exceedance curve with 1-in-10, 1-in-50, 1-in-100 year flood markers.
2. **Term Structure** — annual flood probability over 1–30 years.
3. **PRS Pricing** — six-component pricing breakdown with basis waterfall. Shows base rate, depth threshold components (>0m, >0.5m, >1.0m), and contingent pricing.
4. **Basis Analysis** — gauge spread vs property spread comparison. Basis = gauge spread – property spread (positive = gauge more expensive, expected).

7.4.4 Gauge Hazard Curve

Five-tab panel (1100×750px) loaded from `/api/v1/gauges/{id}/hazard`:

1. **Hazard Curve** — exceedance probabilities for gauge.
2. **Annual Exceedance** — annual probability by depth.
3. **Return Period** — return period curves (1-in- X years).
4. **PRS Pricing** — QuantLib CDS pricing for gauge-level PRS.
5. **Flood Probability** — probability at different thresholds.

7.4.5 Gauge Historical Data

Single-chart panel (700×500px) showing historical daily water levels with flood stage thresholds as horizontal reference lines. Loaded from `/api/v1/gauges/{id}/history`.

7.4.6 Flood Animation

60-frame animation panel (860×620px) showing storm progression. Loaded from `/api/v1/propertyts/animate/{id}`:

- Storm selector dropdown for all available storms.
- Play/Pause controls with speed adjustment (1x/2x/5x).
- Scrubber bar to jump to any hour.
- Gauge markers change colour by water level (green → red).
- Expanding translucent circles from flooded gauges.
- Property markers appear when flood front arrives.
- Stats bar updates per frame: hour, gauges flooded, properties flooded.

7.4.7 Mortgage Detail Panel

Single-section panel (600×520px) loaded from `/api/v1/properties/{id}/mortgage`. Displays five sections: Loan Summary, Financial Terms, Current Status, Borrower, and Risk Assessment.

7.4.8 PDF Report Viewers

Embedded iframe panels (850px wide, 90% viewport height) for viewing generated PDF reports. Include download and print buttons.

7.5 Notification System

Toast-style notifications provide user feedback for all actions:

- **Info** (blue) — general information
- **Success** (green) — operation completed
- **Warning** (orange) — caution
- **Error** (red) — operation failed
- **Loading** (grey) — operation in progress

Auto-dismiss after 5 seconds. Maximum 5 visible simultaneously. Configurable position (default: top-right).

7.6 Backend Communication

Frontend panels communicate with the Flask API via the `window._BACKEND_CONFIG` object, which stores the server URL and endpoint mappings. The backend handler (`src/static/js/backend-handler.js`) manages fetch requests with CORS support and 30-second timeout.

8 PDF Reports

Reports are generated via the API (POST endpoints) or via context menu actions in the interactive map. All reports use ReportLab and follow a consistent branded template.

8.1 Property Reports

15-page comprehensive property analysis:

1. Title and overview
2. Location (map, coordinates, flood zone)
3. Property attributes (type, construction, value)
4. Construction details (materials, age, storeys)
5. Risk assessment (flood depth, damage analysis)
6. Financial (valuation factors, insurance)
7. Flood protection and insurance
8. Energy and EPC rating
9. Transaction history
10. Timeline
11. Mortgage overview (three sub-pages: details, costs, regulatory)

12. Current status
13. Risk analysis
14. Borrower profile
15. Data summary

Generated by `PropertyReportGenerator` in `src/reports/property/`.

8.2 Gauge Reports

11-page gauge analysis:

1. Title and overview
2. Sensor details (type, owner, certification)
3. Location (map, coordinates, ground elevation)
4. Measurements (frequency, method, transmission)
5. Flood stages (alert/warning/severe thresholds)
6. Risk assessment (flood probability, frequency)
7. Data summary
8. Flood history (past events, timeline)
9. Hazard curves (exceedance probability charts)
10. PRS pricing (credit swap valuation)
11. Current risk assessment

Generated by `GaugeReportGenerator` in `src/reports/gauge/`.

8.3 Risk Portfolio Reports

7-page portfolio-level analysis:

1. Title page
2. Executive summary (key metrics, highlights)
3. Portfolio overview (property and gauge distribution)
4. Risk analysis (aggregate flood risk, concentration)
5. Mortgage analysis (LTV distribution, vulnerability)
6. Property details (ranked risk list)
7. Appendix (technical details, methodology)

Generated by `RiskReportGenerator` in `src/reports/risk/`.

8.4 PRS Trade Confirmations

Generated when a PRS trade is committed via `POST /api/v1/prs/commit`. Stored in `data/output/prs/` as both JSON (CDM record) and PDF (trade confirmation). Downloadable via `GET /api/v1/prs/trades/<swap.i`

9 Analytical Models Reference

All analytical functions are centralised in `src/models/`. Portfolio generators and visualisation components are thin wrappers delegating to these models.

9.1 Hazard Models

Table 12: Hazard model modules

Module	Purpose
hazard/gev.py	GEV fitting, exceedance, return levels, term structure
hazard/response_model.py	Gauge response to storms (spatial decay)
hazard/builder.py	End-to-end hazard curve construction
hazard/pricing.py	QuantLib CDS pricing (optional)
hazard/prs_analytical.py	Analytical PRS pricing, basis waterfall
hazard/io.py	Hazard curve file I/O

9.2 Flood Risk Models

Table 13: Flood risk model modules

Module	Purpose
floodrisk/depth_damage.py	Depth-damage curves (UK-calibrated)
floodrisk/spatial.py	Haversine, IDW, KDTree, correlation
floodrisk/velocity.py	Manning's equation, travel time, attenuation
floodrisk/risk_analytics.py	Monte Carlo VaR/ES, HHI, clustering
floodrisk/flood_risk_model.py	Unified flood risk pipeline

9.3 Valuation and Risk Models

Table 14: Valuation, risk, and pricing modules

Module	Purpose
valuation/property_value.py	Multi-factor property valuation
valuation/insurance.py	Insurance premium calculation
risk/risk_assessor.py	Risk scoring, vulnerability, VaR
mortgage_pricer.py	Amortisation, credit spread, expected loss
prs_with_hazard_curve.py	PRS pricing via QuantLib CDS

9.4 Statistics and Intensity Models

Table 15: Statistics, intensity, and storm-gauge modules

Module	Purpose
statistics/timeseries.py	Percentiles, annual maxima, exceedances
statistics/synthetic.py	Stochastic water level generation
intensity/distribution.py	Hybrid Normal-Pareto storm intensity
stormgauge/forward_model.py	Storm-gauge spatial decay, intensity-to-level

9.5 Model Documentation

Detailed mathematical specifications with sensitivity analysis are available as separate PDFs in `docs/models/`. Each model document follows the MKM bank validation protocol with sections for executive summary, purpose, mathematical framework, inputs, implementation, sensitivity, validation, and limitations.

10 Docker Deployment

10.1 Container Image

The Dockerfile builds a production image based on `python:3.13-slim` with GDAL system libraries:

```
1 # Build image
2 docker build -t physicalrisk .
3
4 # Run container
5 docker run -p 5001:5001 -v ./data:/app/data physicalrisk
```

Listing 7: Docker build and run

The container exposes port 5001 and runs `python app.py server --host 0.0.0.0 --port 5001`.

10.2 Docker Compose

```
1 docker-compose up           # Start service
2 docker-compose up --build   # Rebuild and start
3 docker-compose down         # Stop service
4 docker-compose logs -f      # Stream logs
```

Listing 8: Docker Compose usage

Configuration:

- Port mapping: 5001:5001
- Volume: `./data:/app/data` (persists data outside container)
- Environment: `FLASK_ENV=production, CATCHMENT_ID=thames`
- Restart policy: `unless-stopped`

11 Testing

11.1 Test Architecture

The platform has comprehensive test coverage:

- **518 Python tests** across `tests/` (6 skipped)
- **51 JavaScript tests** in `tests/js/`

11.2 Running Tests

```
1 # All Python tests
2 python -m pytest tests/ -v
3
4 # Specific module
```

```
5 python -m pytest tests/models/test_mortgage_pricer.py -v
6
7 # JavaScript tests (requires Node.js)
8 npx jest
9
10 # With coverage
11 python -m pytest tests/ --cov=src --cov-report=html
```

Listing 9: Test execution

11.3 Test Conventions

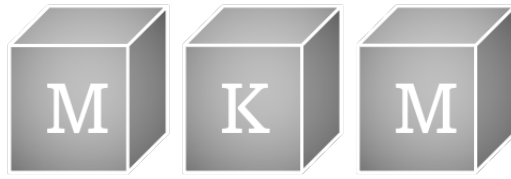
- Tests use `tmp_path` fixture for file isolation.
- Random output is tested with range assertions, not exact values.
- `tests/port/conftest.py` provides shared fixtures for random modules, sample locations, and synthetic data.
- `tests/js/` uses Jest for frontend component testing (notifications, backend handler, context menus).
- `tests/reports/test_pdf_generation.py` has 28 tests for PDF report generation.

Appendix A: Data File Reference

File / Directory	Format	Key Fields
gauge.json	JSON array	GaugeID, lat/lon, elevation, flood stages
property.json	JSON array	PropertyID, lat/lon, elevation, floor level, value
mortgage.json	JSON array	MortgageID, PropertyID, loan, rate, LTV, income
storms.json	JSON array	storm_id, intensity, duration, spatial extent
counterparty.json	JSON array	entity ID, type, credit rating, limits
gaugehc.json	JSON array	gauge_id, GEV params, hazard points, PRS spreads
propertyhc.json	JSON array	property_id, GEV params, PRS pricing, basis
gaugets/GAUGE-*.json	Per-gauge	flood_simulation (hourly readings), storm_responses
gaugehd/GAUGE-*.json	Per-gauge	daily_observations, statistics, percentiles
propertyts/PROP-*.json	Per-property	flood_events (depth, damage, readings)
output/property/*.pdf	PDF	Property analysis reports (15 pages)
output/gauge/*.pdf	PDF	Gauge analysis reports (11 pages)
output/prs/*.json	JSON	PRS trade CDM records
output/prs/*.pdf	PDF	PRS trade confirmations
results/*.html	HTML	Interactive Folium visualisation

12 Document History

Date	Version	Description	Author
14-Feb-2026	1.0	Initial user guide	D. Kelly



Mortgage Pricing Model

Amortisation, Credit Risk, and Flood-Adjusted Expected Loss

MKM Research Labs

Version 1.0 — 13-February-2026

David K Kelly

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1 Executive Summary

This document provides the complete mathematical specification, implementation details, and validation framework for the MKM Mortgage Pricing Model. The model prices individual residential mortgages on a risk-adjusted basis by combining standard amortisation mechanics with a credit-risk overlay derived from borrower affordability, loan-to-value (LTV) ratios, and property-level flood hazard rates.

The core pricing engine computes the present value of expected cashflows net of expected losses, where default probabilities are implied from a hazard-rate framework calibrated to affordability-based credit spreads. Loss given default (LGD) is determined by the shortfall between the outstanding balance and the recoverable property value after application of a recovery haircut.

Key outputs: Fair value of mortgage, credit spread, discount to par, expected loss profile, and survival probability term structure.

Key risk: The model is a downstream consumer of the property hazard curve and PRS pricing models. Flood-adjusted hazard rates feed directly into the survival probability calculation; mis-specification of upstream flood frequencies therefore propagates into mortgage valuation and portfolio-level expected loss aggregates.

2 Model Purpose and Scope

2.1 Purpose

The Mortgage Pricing Model serves three primary functions within the MKM Physical Risk platform:

- **Individual mortgage valuation** — compute the risk-adjusted fair value of a single residential mortgage, accounting for credit risk and flood exposure.
- **Credit spread determination** — derive an implied credit spread from borrower affordability metrics and loan characteristics, enabling comparison with market-observed spreads.
- **Portfolio-level aggregation** — batch-price a portfolio of mortgages and compute aggregate risk metrics (total expected loss, average credit spread, high-risk counts) for regulatory reporting and risk management.

2.2 Scope

- **In scope:** Amortising mortgage cashflow generation; affordability-based credit spread interpolation; LTV-based risk adjustment; hazard-rate default modelling; LGD calculation; expected loss present value; portfolio aggregation.
- **Out of scope:** Prepayment modelling; interest-rate term structure dynamics; stochastic recovery rates; securitisation tranche structuring; operational risk charges.
- **Upstream dependencies:** Property valuation model (property value), property hazard curve model (flood hazard rates), borrower income data.
- **Downstream consumers:** Portfolio flood risk model, risk reporting engine, PRS pricing model (basis risk analysis).

2.3 Regulatory Context

The model is designed to support compliance with:

- PRA SS1/23 — Model Risk Management principles for internal models.
- IFRS 9 — Expected credit loss estimation for financial instruments.
- EBA Guidelines on loan origination and monitoring — affordability assessment.

3 Mathematical Framework

3.1 Amortisation Schedule

The model assumes a standard fully-amortising repayment mortgage with fixed monthly instalments. Given principal P , monthly interest rate $r = r_{\text{annual}}/12$, and total number of monthly payments $n = T \times 12$ (where T is the term in years), the monthly payment M is:

$$M = \frac{P r (1 + r)^n}{(1 + r)^n - 1} \quad (1)$$

In the degenerate case $r = 0$, the payment reduces to $M = P/n$.

The outstanding balance at the end of period i is computed recursively:

$$I_i = B_{i-1} \cdot r \quad (2)$$

$$\Pi_i = M - I_i \quad (3)$$

$$B_i = \max(0, B_{i-1} - \Pi_i) \quad (4)$$

where $B_0 = P$ is the initial loan amount, I_i is the interest component of payment i , and Π_i is the principal repayment component.

For mortgages with a remaining term shorter than the original maturity, the payment is recalculated over the remaining number of periods $n' = T_{\text{current}} \times 12$ using the current outstanding balance:

$$M = \frac{B_0 \cdot r}{1 - (1 + r)^{-n'}} \quad (5)$$

3.2 Credit Spread Model

The credit spread is derived from the borrower's affordability ratio, which measures the proportion of after-tax income consumed by total housing costs.

3.2.1 Affordability Ratio

$$\alpha = \frac{A_{\text{mortgage}} + \iota \cdot V}{Y \cdot (1 - \tau)} \quad (6)$$

where:

- $A_{\text{mortgage}} = 12 \cdot M$ is the annual mortgage payment,
- ι is the annual insurance rate (as a proportion of property value V),
- Y is the gross annual income,
- τ is the income tax rate (default 20%).

The affordability ratio is clamped to the interval $[0.1, 1.0]$ before interpolation.

3.2.2 Base Credit Spread Interpolation

The base credit spread $s_{\text{base}}(\alpha)$ is obtained via linear interpolation over a set of ten calibration nodes:

Table 1: Credit spread calibration schedule

α	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.00
s_{base} (bps)	50	100	200	300	500	800	1200	1800	2500	3500

The interpolation uses `scipy.interpolate.interp1d` with `kind='linear'` and `fill_value='extrapolate'`, permitting extrapolation beyond the boundary nodes. The spread schedule is monotonically increasing and convex, reflecting the non-linear increase in default risk as affordability deteriorates.

3.2.3 Term Structure Adjustment

A multiplicative term factor adjusts the base spread for the position of the remaining term relative to the midpoint of the original maturity:

$$\phi = 1 + \frac{T_{\text{current}} - T_{\text{original}}/2}{100} \quad (7)$$

The final credit spread is:

$$s = \max(0.001, s_{\text{base}}(\alpha) \cdot \phi) \quad (8)$$

enforcing a floor of 10 basis points. For mortgages in the first half of their original term, $\phi < 1$ and the spread is reduced; for those in the second half, $\phi > 1$ and the spread is increased, reflecting the empirical observation that default risk is concentrated in the later amortisation periods when equity cushions are thinner relative to remaining obligations.

3.2.4 Zero-Income Fallback

When the gross annual income is zero or negative (indicating missing or erroneous data), the model assigns a default credit spread of 15% (1500 basis points), consistent with a distressed borrower assumption.

3.3 Loan-to-Value Risk Adjustment

An LTV-based multiplicative risk adjustment factor ρ is applied to capture the additional risk from high-leverage loans:

$$\rho(\text{LTV}) = \begin{cases} 1.0 & \text{if } \text{LTV} \leq 0.80 \\ 1.1 & \text{if } 0.80 < \text{LTV} \leq 0.90 \\ 1.3 & \text{if } 0.90 < \text{LTV} \leq 0.95 \\ 1.5 & \text{if } \text{LTV} > 0.95 \end{cases} \quad (9)$$

where:

$$\text{LTV} = \frac{B_0}{V} \quad (10)$$

The final credit spread after LTV adjustment is:

$$s_{\text{adj}} = s \cdot \rho(\text{LTV}) \quad (11)$$

The step-function form is a deliberate simplification reflecting regulatory LTV bands commonly used in UK mortgage lending (the 80%, 90%, and 95% thresholds correspond to standard product tiers).

3.4 Flood Risk Adjustment

A categorical flood risk multiplier ψ is applied to the credit spread after LTV adjustment. The multiplier captures the additional default risk arising from flood exposure at the property level. Calibration is derived from the `RiskAssessor` value-at-risk percentages (1%, 5%, 15%, 35%, 60%), scaled into spread adjustment factors:

$$\psi(\text{flood risk}) = \begin{cases} 1.00 & \text{Very Low} \\ 1.05 & \text{Low} \\ 1.20 & \text{Medium} \\ 1.40 & \text{High} \\ 1.75 & \text{Very High} \end{cases} \quad (12)$$

The final credit spread after all risk adjustments is:

$$s_{\text{final}} = s_{\text{base}}(\alpha) \cdot \phi \cdot \rho(\text{LTV}) \cdot \psi(\text{flood risk}) \quad (13)$$

where ϕ is the term factor (Equation (7)), ρ is the LTV factor (Equation (9)), and ψ is the flood risk factor. When no flood risk category is provided, $\psi = 1.0$.

3.5 Default and Survival Probabilities

The model employs a reduced-form hazard-rate framework in which the instantaneous default intensity is derived from the credit spread. For each monthly period i , the conditional default probability (hazard rate) is:

$$h_i = 1 - \exp\left(-\frac{s_i}{12}\right) \quad (14)$$

where s_i is the credit spread applicable to period i . In the current implementation, s_i is constant across all periods (flat term structure), but the framework supports period-varying spreads to incorporate flood-adjusted hazard rates from the upstream property hazard curve model.

The survival probability to the end of period i is computed recursively:

$$S_0 = 1 \quad (15)$$

$$S_i = S_{i-1} \cdot (1 - h_i) \quad (16)$$

The marginal (unconditional) probability of default occurring in period i is:

$$PD_i = \max(0, S_{i-1} - S_i) \quad (17)$$

3.6 Loss Given Default

The loss given default in period i is the shortfall between the outstanding balance and the recoverable value of the property:

$$LGD_i = \max(0, B_i - (1 - \delta) \cdot V) \quad (18)$$

where $\delta \in [0, 0.95]$ is the recovery haircut, representing the proportional loss of property value in a forced-sale scenario. The recovery value $(1 - \delta) \cdot V$ is held constant across the life of the mortgage (i.e. no property price appreciation or depreciation is modelled). A haircut cap of 95% is enforced to prevent near-total-loss assumptions.

3.7 Expected Cashflows and Expected Losses

For each period i , the expected cashflow comprises two components:

$$CF_i^{\text{exp}} = M \cdot S_i + (B_{i-1} - LGD_i) \cdot PD_i \quad (19)$$

The first term is the full monthly payment received with probability equal to the survival probability. The second term is the partial recovery received in the event of default: the lender recovers $B_{i-1} - LGD_i$ (i.e. the outstanding balance less the loss) multiplied by the probability of default in that period.

The expected loss for period i is:

$$EL_i = LGD_i \cdot PD_i \quad (20)$$

3.8 Present Value and Fair Value

All cashflows and losses are discounted at the contractual mortgage rate. The discount factor for period i is:

$$D_i = \left(1 + \frac{r_{\text{annual}}}{12}\right)^{-i} \quad (21)$$

The present value of expected cashflows and expected losses are:

$$PV_{\text{CF}} = \sum_{i=1}^n CF_i^{\text{exp}} \cdot D_i \quad (22)$$

$$PV_{\text{EL}} = \sum_{i=1}^n EL_i \cdot D_i \quad (23)$$

The risk-adjusted fair value of the mortgage is:

$$FV = PV_{\text{CF}} - PV_{\text{EL}} \quad (24)$$

The discount to par quantifies the credit value adjustment:

$$\Delta_{\text{par}} = B_0 - \text{FV} \quad (25)$$

$$\Delta_{\%} = \frac{\Delta_{\text{par}}}{B_0} \times 100 \quad (26)$$

3.9 Portfolio Aggregation

For a portfolio of N mortgages, the following aggregate metrics are computed from the set of individual pricing results $\{R_1, R_2, \dots, R_N\}$ (excluding any results flagged with pricing errors):

$$\text{Total Value} = \sum_{j=1}^N \text{FV}_j \quad (27)$$

$$\overline{\text{FV}} = \frac{1}{N} \sum_{j=1}^N \text{FV}_j \quad (28)$$

$$\bar{s} = \frac{1}{N} \sum_{j=1}^N s_j \quad (29)$$

$$\tilde{s} = \text{median}(\{s_j\}_{j=1}^N) \quad (30)$$

$$\overline{\Delta_{\%}} = \frac{1}{N} \sum_{j=1}^N \Delta_{\%,j} \quad (31)$$

$$\text{Total Discount} = \sum_{j=1}^N \Delta_{\text{par},j} \quad (32)$$

$$\overline{\text{LTV}} = \frac{1}{N} \sum_{j=1}^N \text{LTV}_j \quad (33)$$

$$N_{\text{high-risk}} = |\{j : s_j > 0.10\}| \quad (34)$$

A mortgage is classified as “high-risk” when its credit spread exceeds 1000 basis points (10%). The error count tracks mortgages for which the pricing engine returned a fallback result due to exceptions.

4 Input Parameters and Calibration

4.1 Model Inputs

Table 2 summarises the input parameters required by the pricing engine.

Table 2: Input parameters for the Mortgage Pricing Model

Parameter	Symbol	Units	Description
Loan amount	B_0	GBP	Outstanding mortgage balance
Property value	V	GBP	Current market value of property
Interest rate	r_{annual}	Decimal	Annual contractual interest rate
Original maturity	T_{original}	Years	Original mortgage term
Current term	T_{current}	Years	Remaining term to maturity
Gross annual income	Y	GBP	Borrower's gross annual income
Insurance rate	ι	Decimal	Annual insurance as proportion of V
Recovery haircut	δ	Decimal	Forced-sale loss proportion
Tax rate	τ	Decimal	Income tax rate (default 0.20)

4.2 Input Validation and Bounds

The model applies the following validation rules to all inputs before computation:

Table 3: Input validation bounds

Parameter	Constraint	Rationale
B_0	≥ 1	Prevents division by zero in discount calculation
V	≥ 1	Prevents division by zero in LTV calculation
Y	≥ 1	See zero-income fallback (Section 3.2)
r_{annual}	≥ 0.001	Floor of 0.1% to ensure meaningful discounting
ι	≥ 0	Insurance rate cannot be negative
T_{original}	≥ 1	Minimum one-year term
T_{current}	$[0.5, T_{\text{original}}]$	Minimum six months; cannot exceed original
δ	$[0, 0.95]$	Recovery haircut capped at 95%

4.3 Calibration

4.3.1 Credit Spread Schedule

The credit spread calibration nodes (Table 1) are derived from historical UK mortgage default rates segmented by debt-service ratio, sourced from Council of Mortgage Lenders (CML) data and cross-referenced with Bank of England Financial Stability Reports. The schedule is reviewed annually and updated when cumulative default experience deviates by more than 20% from implied spreads.

4.3.2 Recovery Haircut

The default recovery haircut should be calibrated to observed forced-sale discounts in the relevant property market. For UK residential property, a typical range is $\delta \in [0.15, 0.35]$, reflecting 15–35% discount relative to open-market value. The example configuration uses $\delta = 0.25$ (25% haircut).

4.3.3 Tax Rate

The default tax rate of $\tau = 0.20$ corresponds to the UK basic rate of income tax. This parameter should be overridden for higher-rate taxpayers or when modelling portfolios in different tax jurisdictions.

5 Implementation Details

5.1 Software Architecture

The model is implemented in the Python module `src/models/mortgage_pricer.py` and comprises:

- `MortgagePricer` class — pricing engine initialised with an optional tax rate (default 20%). The credit spread interpolation function is constructed at initialisation.
- `calculate_monthly_payment()` — static method; computes M per Equation (1). Takes principal, annual interest rate (decimal), and term in years.
- `calculate_total_cost()` — static method; computes $M \times n$ (undiscounted).
- `calculate_credit_spread()` — implements the affordability-based spread model (Section 3.2).
- `calculate_loan_to_value_impact()` — returns $\rho(\text{LTV})$ per Equation (9). Called by `price_mortgage()` to adjust the credit spread (Equation (11)).
- `price_mortgage()` — full pricing pipeline returning a dictionary of results including fair value, cashflow arrays, and risk metrics. Errors propagate to the caller rather than returning fallback values.
- `batch_price_mortgages()` — iterates over a portfolio list, calling `price_mortgage()` for each entry with error isolation.
- `calculate_portfolio_metrics()` — module-level function computing aggregate statistics from batch results.

5.2 Dependencies

- `numpy` — array operations for balance, hazard, and cashflow vectors.
- `scipy.interpolate.interp1d` — linear interpolation of the credit spread schedule.

5.3 Numerical Considerations

1. **Vectorisation:** The outstanding balance, hazard rates, survival probabilities, and LGD arrays are pre-allocated as NumPy arrays of length $n + 1$. Expected cashflows and losses use arrays of length n . Discount factors are computed as a single vectorised operation over the period index array.
2. **Balance recursion:** The amortisation schedule is computed via a sequential loop (period i depends on B_{i-1}). This is an inherently serial computation; however, for typical mortgage terms (up to 360 periods) the overhead is negligible.
3. **Discount rate:** The contractual mortgage rate is used as the discount rate. This means the risk-free present value of the contractual cashflows equals the principal at inception (par). The discount to par therefore represents the credit value adjustment.
4. **Error handling:** The `price_mortgage()` method performs input validation and clamping but allows exceptions to propagate to the caller. The `batch_price_mortgages()` method provides error isolation at the portfolio level, recording errors per mortgage without halting the batch.

5.4 Computational Complexity

For a single mortgage with n monthly periods:

- Time complexity: $O(n)$ — single pass through the amortisation schedule and a single vectorised discount-factor computation.
- Space complexity: $O(n)$ — storage for balance, hazard, survival, LGD, cashflow, and loss arrays.
- For a portfolio of N mortgages: $O(N \cdot n)$ total, fully parallelisable across mortgages (currently sequential).

6 Sensitivity Analysis

The following parameters have the greatest influence on the model's fair value output.

Table 4: LTV risk multiplier by loan-to-value ratio (property value = £300,000).

LTV	Loan (£)	Risk Multiplier
50%	150,000.0	1.00
60%	180,000.0	1.00
70%	210,000.0	1.00
80%	240,000.0	1.00
85%	255,000.0	1.10
90%	270,000.0	1.10
95%	285,000.0	1.30
100%	300,000.0	1.50

Table 5: Mortgage value (£) by interest rate and insurance rate (loan=£200k, income=£50k, 20yr remaining).

Interest rate	Ins=0.2%	Ins=0.5%	Ins=1.0%	Ins=2.0%
2%	199,937.0	199,931.0	199,922.0	199,895.0
3%	199,898.0	199,890.0	199,875.0	199,831.0
4%	199,855.0	199,844.0	199,814.0	199,757.0
5%	199,802.0	199,780.0	199,746.0	199,671.0
6%	199,733.0	199,709.0	199,670.0	199,578.0

Table 6: Credit spread and mortgage value by gross annual income (loan=£200k, rate=4%, 20yr remaining).

Income	Spread (%)	Mortgage Value (£)
£30,000	11.5	199,631.0
£40,000	5.42	199,769.0
£50,000	3.25	199,844.0
£60,000	2.52	199,874.0
£80,000	1.62	199,915.0
£100,000	1.08	199,941.0

Table 7: Credit spread and mortgage value by flood risk category (loan=£200k, rate=4%, income=£50k).

Flood Risk	Multiplier	Spread (%)	Mortgage Value (£)
Very Low	1.00	3.25	199,844.0
Low	1.05	3.41	199,838.0
Medium	1.20	3.90	199,820.0
High	1.40	4.55	199,797.0
Very High	1.75	5.68	199,761.0

6.1 Interest Rate Sensitivity

The monthly payment M is convex in r , and the discount factors D_i decrease with r . An increase in the contractual rate simultaneously raises cashflows and increases discounting. The net effect on fair value depends on the magnitude of the credit spread relative to the base rate.

Expected behaviour:

- A 100 bps increase in r_{annual} increases M by approximately 6–8% for a 25-year term.
- The discount-to-par typically increases with r because higher payments worsen affordability, raising s and hence expected losses.

6.2 Affordability Ratio Sensitivity

The credit spread schedule is highly non-linear: moving from $\alpha = 0.3$ (200 bps) to $\alpha = 0.5$ (500 bps) represents a 150% increase, whilst moving from $\alpha = 0.7$ (1200 bps) to $\alpha = 0.9$ (2500 bps) represents a 108% increase. The model is therefore most sensitive to affordability deterioration in the mid-range $\alpha \in [0.3, 0.6]$, where marginal changes produce the largest absolute spread movements.

6.3 Recovery Haircut Sensitivity

The LGD calculation is directly proportional to δ . For a mortgage at 80% LTV with $V = £500,000$ and $B_0 = £400,000$:

- $\delta = 0.15$: Recovery = £425,000, LGD = 0 (fully covered).
- $\delta = 0.25$: Recovery = £375,000, LGD = £25,000.
- $\delta = 0.40$: Recovery = £300,000, LGD = £100,000.

The sensitivity is amplified at high LTV ratios, where the equity cushion is insufficient to absorb the haircut.

6.4 Term Structure Sensitivity

The term factor ϕ has a modest effect: for a 30-year original term with 25 years remaining, $\phi = 1 + (25 - 15)/100 = 1.10$, a 10% uplift to the base spread. Conversely, a mortgage with only 5 years remaining gives $\phi = 1 + (5 - 15)/100 = 0.90$, a 10% reduction. This factor is intentionally conservative to avoid over-fitting to maturity effects.

7 Validation and Backtesting

7.1 Unit Testing

The model is covered by the test suite in `tests/`. Key test cases include:

- **Monthly payment:** Verified against standard financial calculator results for representative loan parameters (tolerance $< £0.01$).
- **Zero-rate degenerate case:** Confirms $M = P/n$ when $r = 0$.
- **Credit spread bounds:** Verifies the floor at 10 bps and the zero-income fallback at 1500 bps.
- **Balance convergence:** Confirms $B_n = 0$ (within floating-point tolerance) at the end of the amortisation schedule.
- **Survival monotonicity:** Verifies $S_i \geq S_{i+1} \geq 0$ for all i .
- **LTV risk factor:** Confirms correct step-function thresholds.
- **Batch pricing:** Verifies error isolation (one failed mortgage does not prevent others from pricing).
- **Portfolio metrics:** Verifies aggregation formulae against hand-computed values.

7.2 Analytical Benchmarks

7.2.1 Par Recovery

For a risk-free mortgage ($s = 0$, $\delta = 0$), the fair value must equal the loan amount (par). This can be verified analytically: with $h_i = 0$ for all i , we have $S_i = 1$, $PD_i = 0$, and $EL_i = 0$, so $FV = PV_{CF} = B_0$.

7.2.2 Maximum Loss Bound

The total undiscounted expected loss is bounded above by the loan amount:

$$\sum_{i=1}^n EL_i \leq B_0 \quad (35)$$

This holds because $\sum PD_i \leq 1$ and $LGD_i \leq B_i \leq B_0$.

7.3 Backtesting Protocol

For ongoing model validation, the following backtesting programme is recommended:

1. **Monthly:** Compare model-implied credit spreads against market CDS spreads for listed UK mortgage-backed securities.
2. **Quarterly:** Compare cumulative default predictions against observed portfolio arrears data.
3. **Annually:** Re-calibrate the credit spread schedule using updated CML/FCA default statistics. Assess recovery haircut calibration against realised forced-sale outcomes.

8 Model Limitations and Known Weaknesses

1. **Flat credit spread term structure:** The current implementation applies a constant credit spread across all periods. In reality, default risk varies over the life of a mortgage (typically

higher in early years, lower as equity builds). A time-varying spread schedule would improve accuracy.

2. **Deterministic recovery:** The recovery value is fixed at $(1 - \delta) \cdot V$ throughout the mortgage term. This ignores property price volatility, which is material for long-dated mortgages. A stochastic property value model would capture procyclical recovery risk.
3. **No prepayment modelling:** The model assumes no voluntary prepayment. In practice, UK mortgage borrowers frequently remortgage at product maturity (typically every 2–5 years). Prepayment shortens effective duration and reduces expected loss.
4. **Step-function LTV adjustment:** The discrete LTV risk bands (Equation (9)) create discontinuities at the 80%, 90%, and 95% thresholds. A continuous function (e.g. logistic or power-law) would be more robust.
5. **Linear spread interpolation:** The credit spread function uses linear interpolation with boundary clamping. For affordability ratios outside $[0.1, 1.0]$, spreads are clamped to the nearest boundary value (50 bps or 3500 bps respectively).
6. **Single discount rate:** Cashflows are discounted at the contractual mortgage rate rather than a risk-free rate plus spread decomposition. This conflates the time value of money with credit compensation.
7. **No correlation modelling:** Portfolio aggregation treats mortgages as independent. In practice, systemic flood events or economic downturns create correlated defaults. The portfolio expected loss is therefore likely understated in tail scenarios.
8. **Error propagation:** The `price_mortgage()` method raises exceptions on invalid inputs rather than returning fallback values. Downstream consumers must handle exceptions or use `batch_price_mortgage()` which provides per-mortgage error isolation.
9. **Insurance rate held constant:** The model does not account for potential withdrawal of flood insurance cover or premium increases following flood events, which would worsen affordability dynamically.

9 Change History

All material changes to the model methodology, calibration, or implementation are recorded in the document history table. Changes are classified as:

- **Material** — changes to mathematical formulae, calibration data, or model scope that affect output values.
- **Non-material** — documentation corrections, code refactoring without output changes, or commentary updates.

The model owner is responsible for maintaining this log and ensuring that all material changes trigger a re-validation cycle.

10 Document History

Date	Version	Description	Author
13-Feb-2026	1.0	Initial documentation	D. Kelly